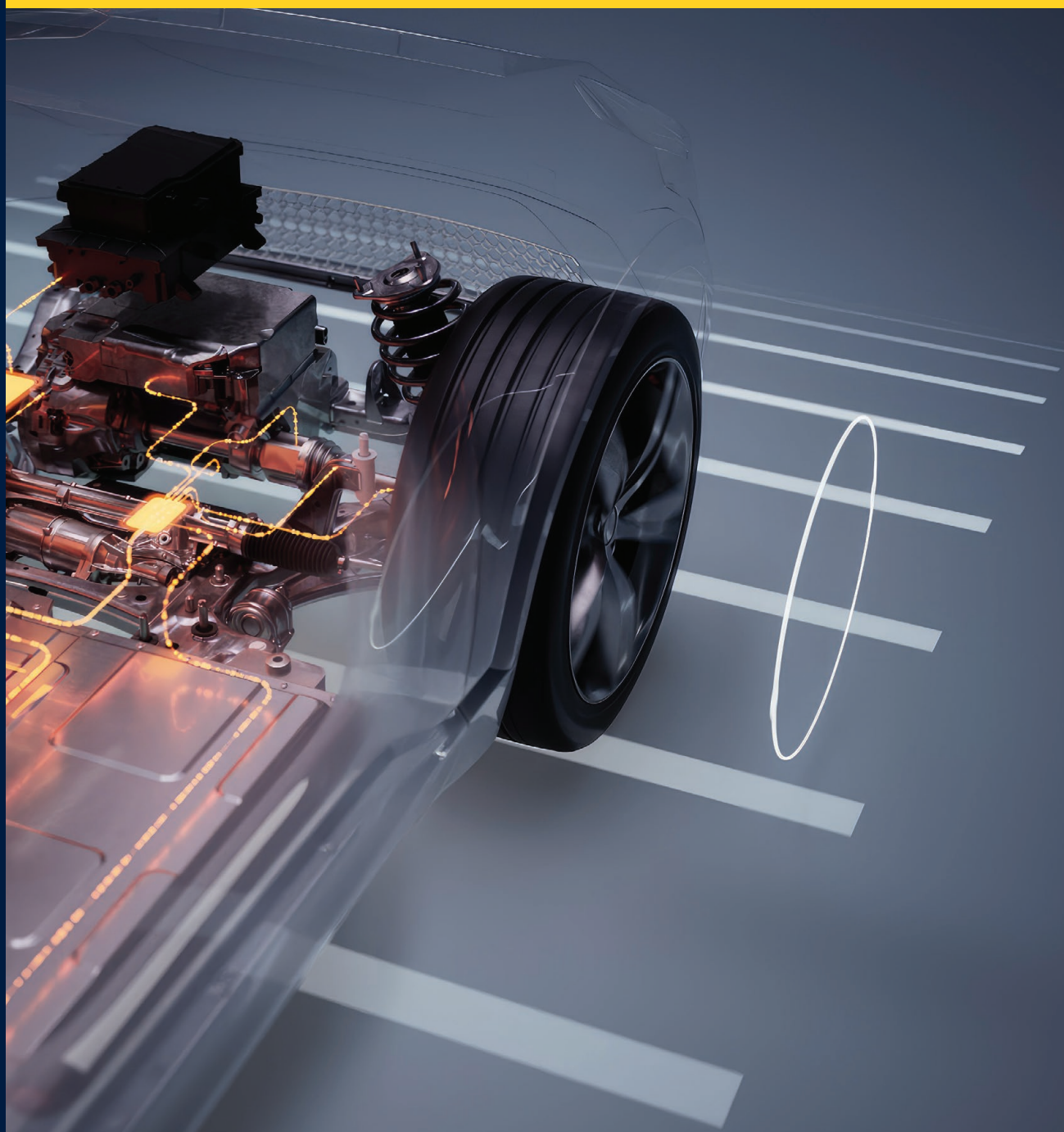




Unleashing the software-defined vehicle revolution with xMemory

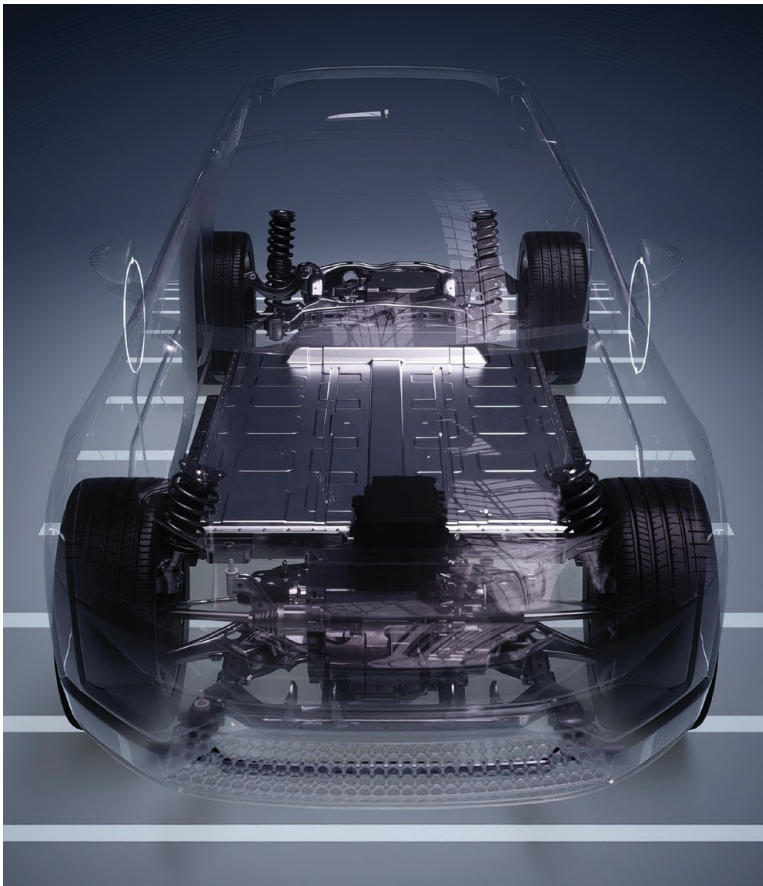


Driving the future: navigating the shift to software-defined vehicles and memory-centric automotive architectures

The automotive industry is undergoing a rapid transformation, driven by the emergence of software-defined vehicles (SDVs) and the acceleration of electrification. This shift is fundamentally reshaping traditional electronic/electrical (E/E) architectures, transitioning from hardware-centric designs to software-driven platforms where innovation and differentiation are critical. This evolution necessitates not only unprecedented processing power but also a reimagined approach to system architecture for each chip in a car.

Differentiation is crucial as it enables companies to:

- **Stand out in a highly competitive market** by offering unique features, designs, or technologies.
- **Meet diverse customer needs and preferences**, enhancing brand loyalty and customer satisfaction.
- **Drive innovation**, such as advanced safety systems, electric powertrains, or connectivity features.
- **Command premium pricing** by providing distinctive value that competitors lack.
- **Adapt to regulatory and environmental challenges** through differentiated solutions like low-emission or autonomous vehicles.



Differentiation now relies on delivering innovative features and functionalities through software and over-the-air (OTA) updates. This transformation is critical for the automakers because it turns vehicles into dynamic platforms offering evolving services. For customers, this means their vehicles continuously improve, gaining new capabilities and retaining value longer. This transformation significantly increases the number of lines of code, directly impacting electronic control units (ECUs) and their embedded microcontrollers (MCUs).

For example, emerging capabilities, such as anomaly detection and virtual sensors, leverage artificial intelligence (AI) to perform real-time tasks close to the sensors. These AI-driven functions require additional memory to store vehicle parameters and input data.

This evolution presents a critical challenge for engineers: balancing the need for increased memory with the time-intensive task of optimizing existing code within constrained memory footprints. For businesses, late-stage changes or the addition of new hardware components to vehicle architectures introduce complexity, increasing operational costs and extending development timelines due to rigorous automotive-grade reliability and safety verification processes. With the pace of automotive manufacturing accelerating, automakers are under increasing pressure to reduce risks that directly impact product differentiation.

This whitepaper explores how **Stellar with xMemory** goes beyond a mere technological advancement—it embodies a strategic imperative for businesses striving to reduce end-product costs, optimize resource allocation, mitigate design risks, and maximize long-term value in today's rapidly evolving market.

Navigating memory bottlenecks in modern automotive systems

The fundamental shift in automotive E/E architectures highlights a critical bottleneck: the limitations of traditional memory solutions. As automotive systems evolve toward software-defined vehicles (SDVs), traditional Flash memory faces technical and scalability challenges. Increasing demands for performance, capacity, and endurance at smaller process nodes call for advanced embedded solutions.

The development of cutting-edge memory technologies like phase-change memory (PCM) is reshaping the landscape. This shift highlights ST's leadership in delivering robust, high-density eNVM optimized for the automotive industry's rigorous requirements.

From Flash to embedded NVM: The scalability imperative

For decades, embedded Flash memory have been critical in automotive Microcontrollers, primarily for code storage and non-volatile data. The increasing demands of SDVs, multiple functions, and AI applications necessitate new chips with smaller process nodes and much greater memory capacity. Flash technology struggles to provide the high performance, density, and endurance required for continuous SW updates at the right cost structure. This necessitates a move towards more advanced embedded non-volatile memory (eNVM) solutions. Three emerging technologies are poised to replace embedded Flash memory: phase-change memory (PCM), resistive RAM (RRAM), and magnetoresistive RAM (MRAM).

After more than a decade of development, ST has successfully transitioned from Flash memory to PCM technology, achieving more than twice the density compared to other technologies. ST has further optimized PCM by integrating it with IDM technologies such as FD-SOI, resulting in superior performance and exceptional robustness being the **first to qualify** to the most stringent temperature grade **Automotive Grade 0**.

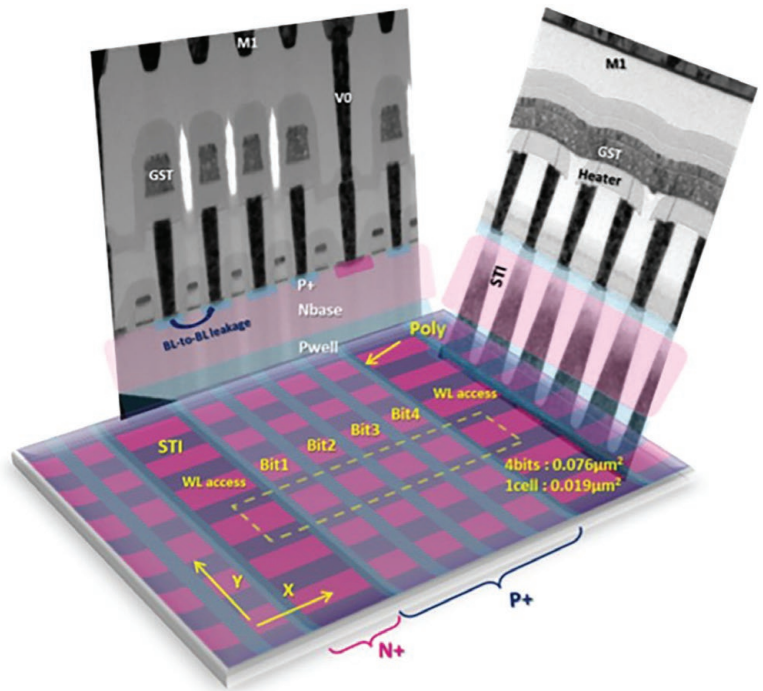


Figure 1: Representation of memory cells in a phase-change memory

Innovating through software: addressing memory and operational challenges

Memory constraints are now a major roadblock for our customers, hindering their ability to innovate while creating significant operational inefficiencies:

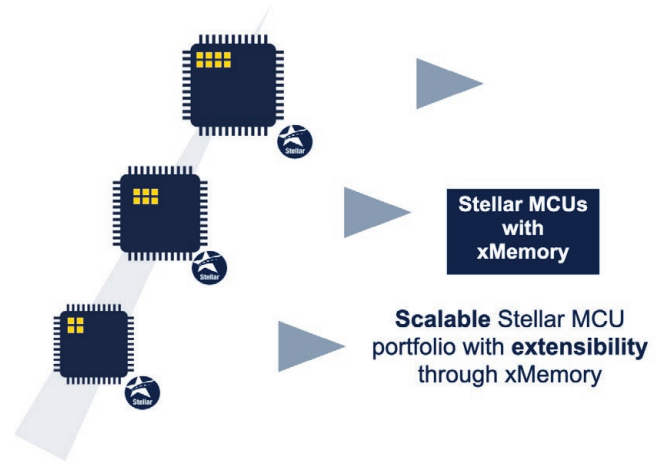
- **More functions per chip:** The increasing number of software functions per MCU inevitably leads to a need for more memory.
- **Difficult to maintain:** A proliferation of components with various memory capacities adds layers of complexity, increasing the potential for integration issues and prolonging validation cycles.
- **Lack of scalability** or limited lifetime: Systems built with insufficient memory capacity face a restricted operational lifetime. As new features are pushed via Over-The-Air (OTA) updates, systems with fully utilized memory may quickly become obsolete, necessitating more costly hardware refreshes.
- **Limited differentiation/innovation:** Ultimately, restricted memory capacity stifles the SDV transition. OEMs are forced to make difficult trade-offs, limiting the scope of new features and functionalities they can deliver, thereby impeding their ability to differentiate themselves through software in a highly competitive market.

Stellar with xMemory

Driving the automotive SDV transition

Introducing **Stellar with xMemory**: the ideal combination of an automotive-compliant, software-defined vehicle (SDV)-optimized system architecture and automotive-qualified phase-change memory (PCM) technology, specifically designed to overcome the memory constraints limiting the software-defined vehicle revolution.

Stellar with xMemory unlocks additional savings and speeds up development with a single part solution that is able to meet multiple projects memory requirements.



Our innovative approach: phase-change memory optimized for automotive

Our distinctive advantage stems from our **Integrated Device Manufacturing (IDM) capability**, meaning xMemory was developed in-house. This end-to-end control over design, process, and manufacturing allows for unparalleled optimization, tighter integration, and faster iteration cycles as well as supply security. Unlike fragmented solutions, our approach ensures that xMemory is purpose-built and perfectly tailored for the rigorous demands of the automotive market.

TECHNICAL DEEP DIVE: REDEFINING MEMORY DENSITY AND PERFORMANCE

Unipolar architecture

A significant technical advantage of phase-change memory (PCM) is the simplification of memory cell design and control circuitry enabled by its unipolar operation. This contrasts with other emerging technologies, such as resistive RAM (RRAM) and magnetoresistive RAM (MRAM), which require current direction switching for write operations. Unipolar operation in PCM means both the "set" (crystallization) and "reset" (amorphization) states, representing binary 0s and 1s, are achieved by applying current pulses of the same voltage polarity but different magnitudes or durations, primarily leveraging Joule heating for phase transitions.

This eliminates the need for complex bidirectional current control in the periphery and allows an optimal area efficiency solution for the bit-cell selector, ultimately contributing to a very dense memory and to a simplified driving schemes.

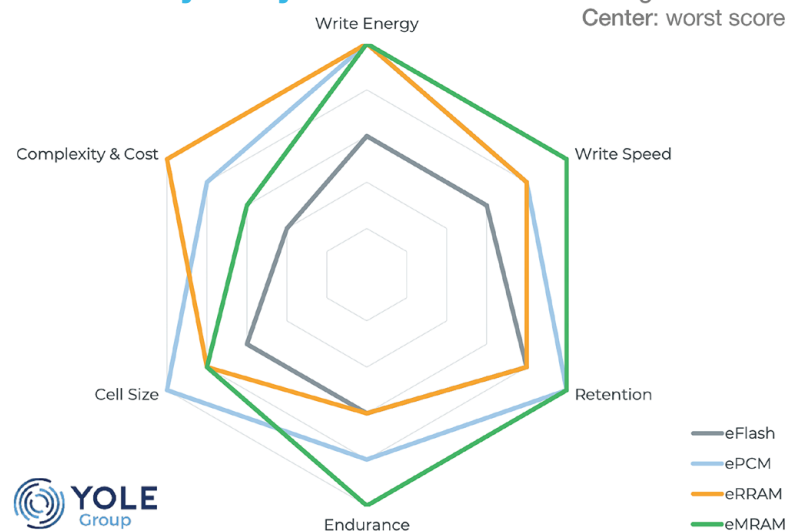
Automotive reliability

PCM demonstrates high endurance, capable of withstanding **250,000 write cycles** at temperatures up to **165°C**. Additionally, the memory can support up to **25 years** of data retention, meeting the stringent requirements for storing automotive code.

Our in-house development has also allowed us to optimize the internal read/write circuitry and control logic, leading to superior characteristics such as better reliability and enhanced support for high temperature robustness.

xMemory is **the first emerging eNVM** that is meeting and **exceeding the most stringent Automotive Grade 0 requirements** for reliability and operational temperature range.

An industry analyst view



Cell size at 20nm-class nodes

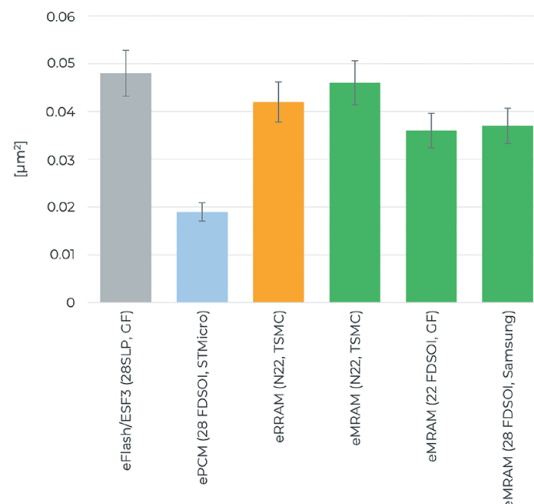


Figure 2: Comparison of embedded memory technologies (Courtesy of the Yole Group)

In its “Emerging Non-Volatile Memory 2025” report, Yole Development has performed a thorough analysis of the embedded and stand-alone NVM technologies, applications, and related markets.

Embedded PCM stands out for its compact cell size, making it particularly well-suited for high-density non-volatile memory integration in advanced microcontrollers. Its inherent robustness and reliable data retention under extended temperature and cycling conditions position it as a strong candidate for automotive-grade applications where long-term stability is critical.

Simone Bertolazzi, PhD, Principal Analyst at Yole Group

The superior density of the NVM lies at the core of xMemory, enabling the extension of the Stellar MCU portfolio to a broader range of applications and products. This translates into greater innovation for customers, while reducing costs and accelerating time to market with a streamlined number of parts (SKUs).

	eFlash		PCM	RRAM		MRAM		
								
Node [nm]	28 (ESF3)	28 (MONOS)	28 & 18 FDSOI	22	22 FDSOI	22	28 FDSOI	22 FDSOI
Cell size [μm^2]	0.043	0.045	0.019 ⁽¹⁾	0.042 ⁽²⁾	0.046 ⁽³⁾	0.046	0.036 ⁽⁴⁾	0.047
Endurance [cycles]	100-300K	10K-250K	250K	10K	10K	100K-1M	1M	1M
Retention	20y (125 °C)	20y (>150 °C)	20y (>150 °C)	10y (125 °C)	10y (125 °C)	20y (150 °C)	10y (125 °C)	20y (150 °C)
Additional Masks	9-16 ⁽⁵⁾		N.A. ⁽⁶⁾	2		2-3		
Concern	High mask adder		Drift	Variability		Magnetic Immunity		
Competitive Edge	Auto Grade-0 & MLC capable	AEC-Q100 Grade 0 High read speed	AEC-Q100 Grade 0	Low cost & complexity		Auto Grade-1 High performance		

Notes: (1) Single bit-per-cell configuration. ST has addressed all critical challenges to achieve Grade-0 qualification, with all designs supporting this configuration. (2) TSMC manufactures RRAM-based automotive products for Infineon at the 28nm node using a process that is exclusive to Infineon (data not disclosed). The cell size is expected to be larger to enhance reliability. (3) GlobalFoundries is developing also a 0.038 μm^2 RRAM cell. (4) Samsung also offers alternative eMRAM solutions at the same node, using larger cell sizes to improve high-temperature retention. (5) Layer count includes adding the High-Voltage (HV) devices to the baseline process. (6) The exact number is not available; it is expected to be slightly higher than for RRAM and MRAM, since additional masks are required for both storage and bipolar elements. Sources: Technical literature, reverse engineering, and interviews conducted by Yole Group.

Table 1: Key technical characteristics of eNVM technologies (Courtesy of the Yole Group)

Solving the memory allocation dilemma

In automotive ECU design, rapid industry changes force product designers to face a critical trade-off: either oversize memory to avoid risks—leading to higher costs and reduced competitiveness—or use constrained memory, risking limited functionality and future adaptability. Stellar with xMemory is set to redefine this paradigm by eliminating this restrictive choice, delivering significant operational and financial efficiencies.

Enhanced flexibility and better cost

Avoid the pitfalls of under-provisioned memory devices and eliminate the need for oversized ones by deploying solutions that provide precisely the right computational resources at launch, along with sufficient memory for future updates.

This approach significantly reduces upfront material costs and improves profit margins.

Streamlined product development & deployment

Engineers can focus on core functionalities without the burden of rigid memory constraints. A single product can support more functions, multiple product lines, and even span several generations.

This accelerates design cycles and allows for more innovative product roadmaps, knowing memory can handle evolving features.

Simplified procurement & inventory

Consolidate your supply chain by reducing the number of distinct parts (SKUs) you manage, leading to higher volume orders, better pricing, and reduced warehousing costs as well as lower obsolescence risk

Stellar MCUs with xMemory have already been adopted by leading customers, proving that they are not just a technological breakthrough but a strategic necessity for businesses seeking to lower end-product costs, optimize resource use, reduce design risks, and maximize long-term value in a fast-changing market.

Case study: Cost savings for Tier-1 supplier

A key Tier-1 supplier faced the challenge of developing a scalable ECU platform that could be adapted for 3 different vehicle applications, each with varying memory requirements, while minimizing inventory complexity and development timelines.

By selecting Stellar with xMemory, the Tier-1 supplier was able to get a single part fitting multiple memory requirements, effectively extending one core device to cater to multiple product variants.

Results

The cost benefits were immediately apparent, as procurement costs - including shipping and storage - can account for up to 20% of the final product cost. This helped them streamline inventory management and procurement.

The ability to reuse a single qualified component across different products drastically accelerated their development cycles and reduced overall qualification costs. Furthermore, the advantages extend beyond product and development cost savings.

For automakers, supply chain disruptions, such as those experienced during the pandemic, can be highly challenging and expensive. A single missing component can halt car production, leading to significantly higher costs.

By choosing Stellar with xMemory, our customer mitigated these risks, ensuring a more resilient and cost-effective production process.

Case study: Integrating new AI functions

One of our top OEM customers wanted to enhance the safety for its autonomous driving solution by integrating AI functions into the companion MCU that verifies power and signal integrity on an ADAS platform. Traditional algorithms previously used to detect anomalies in the system suffered from high false positive and negative rates. To improve accuracy, this OEM sought to additionally integrate AI models excelling at outlier detection. By integrating xMemory, the OEM gained the necessary memory headroom – substantially greater than competitive solutions – enabling support for additional AI functions on a single ECU without any performance degradation.

Results

This enabled the OEM to implement more advanced, memory-intensive AI functions concurrently, leading to a safer autonomous driving solution. The headroom provided by xMemory left the space for future updates and innovations.

Product availability and flexible memory options

Production of **Stellar with xMemory** is scheduled to begin later in 2025 starting with the Stellar P and G series. To meet specific project needs, two main memory configuration options are available:

Fixed memory option

This cost-effective solution is ideal for projects with clear and stable memory needs. It uses the same technology but has a fixed, non-expandable memory size. It's best suited for a single project with a known configuration and when no future upgrades are anticipated.

Flexible memory option

This single-part memory solution is ideal for customers with diverse or uncertain memory needs across multiple projects, allowing for easy expansion up to maximum capacity before production or even on the field. It offers cost savings by enabling reuse and supports longer product lifecycles through continuous software updates.

Ultimately, the flexibility offered with xMemory ensures customers can optimize product and development costs, simplify and accelerate designs, and confidently futureproof their innovations, also leveraging the scalability of the Stellar MCU portfolio.



Conclusion

The automotive industry stands at the forefront of a software-driven transformation. To unlock the full potential of software-defined vehicles—with dynamic features, advanced AI capabilities, and seamless over-the-air updates – a new foundation for memory technology is not just advantageous but essential.

xMemory delivers on this imperative by driving two complementary innovations:

Technological innovation

Developed in-house at ST, xMemory offers unprecedented density - over twice that of competing solutions - and allows larger memory capacity for innovation. This breakthrough enables superior performance, streamlined system architectures, and future-proof designs.

Business innovation

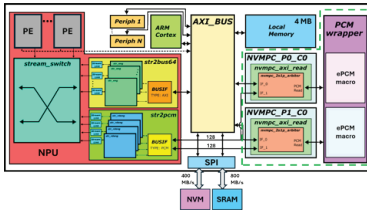
By reducing component count, accelerating qualification processes, maximizing reuse, and enabling more complex features, Stellar MCUs extend to more projects and deliver substantial cost savings, faster time-to-market, and a clear competitive advantage for automakers striving for differentiation.

Getting started



For the technical community

Ready for a deeper dive? Visit our [phase-change memory \(PCM\) technology and innovation page](#) to find out more about how PCM is manufactured and its applications.



Best-in-Class Density Single-Ended ePCM Memory Array for Weight Storage in Edge-AI Applications

In this [conference paper](#) (IEEE subscription required), Andrea Redaelli, PCM Process Architecture Technical Director and ST company fellow, highlights the enablement of a high-density memory array in a System on Chip (SoC) as a great opportunity to overcome the Von Neumann bottleneck in computation, particularly for edge applications.



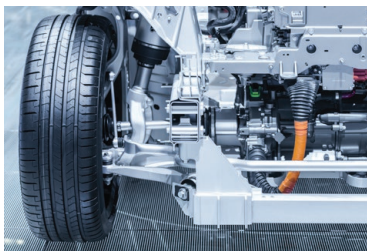
Emerging Non-Volatile Memory 2025 report (Yole Group)

The [2025 update of the “Emerging Non-Volatile Memory” report](#) provides a deep analysis of emerging non-volatile memory markets and technologies in both stand-alone and embedded forms.



For developers

What could you create if you had more memory than ever before? Start developing on Stellar today using the [Stellar development ecosystem](#) and unlock the full potential of xMemory.



For automakers

What advanced features will you unlock today and in the future with the unparalleled capacity and flexibility of xMemory? Check the [ST's Scalable Automotive Stellar MCUs with xMemory page](#) and the [“Moor Insights & Strategy” research paper](#).

Let's shape the future of mobility together with www.st.com/stellar



For business leaders and analysts

Let's collaborate to build a detailed business case tailored to your specific needs, showcasing the significant ROI and strategic advantages that xMemory can deliver to your organization.

Please [contact your nearest STMicroelectronics local sales office](#) to get started.

At STMicroelectronics
we create
technology that
starts with You



Order code: **WP2508XMEMORY**

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